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The Interchangeability of the Cross-Platform Data in the Deep Learning-based Land Cover Classification Methodology

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Abstract. This study employs a validated airborne Light Detection and Ranging (LiDAR) bathymetry system (ALB) and a UAV-borne Green LiDAR System (GLS) for cross-platform analysis of land cover classification (LCC). Furthermore, the study aims to visualize LiDAR data using high-contrast colour scales and improve the accuracy of land cover classification methods through image fusion techniques. If high-resolution aerial imagery is available, it needs to be downscaled to match the resolution of the low-resolution point clouds. The interchangeability of cross-platform data was assessed by comparing the interchangeability, which measures the absolute difference in overall accuracy (OA) or macro-F1 by comparing the cross-platform inter prediction. It is important to note that relying solely on aerial photographs is inadequate for achieving precise labelling, particularly under limited sunlight conditions that can result in misclassification. In such cases, LiDAR plays a crucial role in facilitating target recognition.

Keywords: Airborne LiDAR Bathymetry, Cross-platform, Deep Learning, Green LiDAR System, Riverine Land Cover Classification

1 Introduction

If the data collected by different platforms, the impacts on the LCC mapping results derived by this operation has not been demonstrated yet. To observe the impact of cross-platform on LCC in this study as most as possible, the resolution and data styles (i.e., digital imagery, LiDAR) of ALB [1] and GLS in mutual-prediction are chosen as same. On the other side, because of the data limitation, instead of totally same season, the similar season-related ALB and GLS dataset are selected. Eventually, the comparisons in this research obtained from cross-platform mutual-predictions can facilitate understanding of cross-platform data features, and determine a reasonable method to retain the robustness using deep learning method with cross-platform data in LCC mapping producing.

2 Data processing

Figure 1 depicts the processing of producing imagery input data for this study [2]. It includes three types of data, i.e., LR-DI (low-resolution digital images derived from high-resolution digital images), LiDAR-I (images derived from LiDAR-based point cloud using a high-contrast color bar), and Image Fusion (images created by superimposing the above images).

3 Work flow

To extract features from the input RGB-based imagery (i.e., LR-DI, LiDAR-I, Image Fusion), the processing of training and inference is represented as following: 1. Trimming the raw data from cross-platform input dataset with preprocessing module to attain imagery-based input data (320 pixel × 320 pixel), then training the input data with DeepLabV3+ model to achieve trained model; 2. Predicting the imagery-based input data with trained model. To end with attaining the results (i.e., OA and

Macro-F1 from mutual-prediction, separately); 3. Comparing the averaged and absolute difference value for confirming interchangeability of cross-platform dataset.

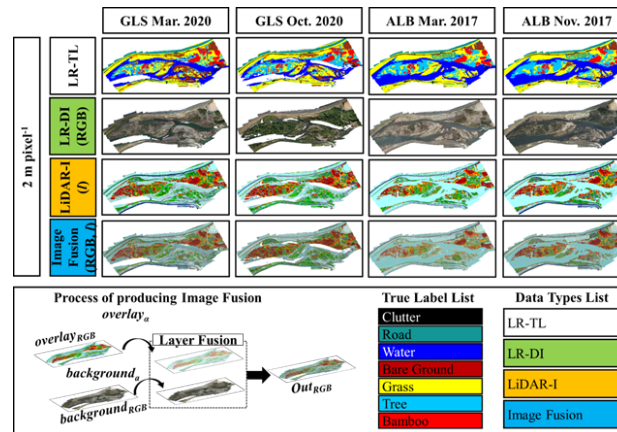


Figure 51 2 m pixel⁻¹ Imagery-based input (LR-TL, LR-DI, LiDAR-I, Image Fusion), Image Fusion processing, True Label List and Data Types.

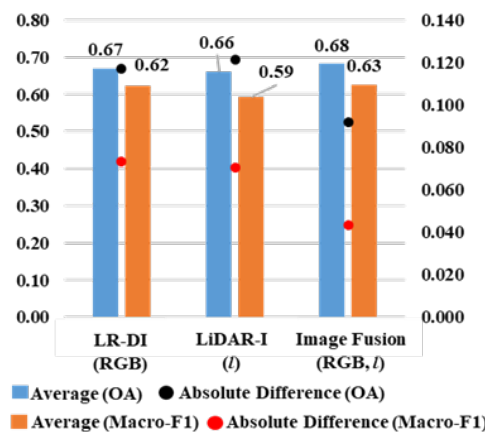


Figure 2 Comparison of data style-based averaged 2m pixel⁻¹ resolution cross-platform interchangeability; Left vertical axis: the reference of OA and Macro-F1 value; Right vertical axis: the reference of Absolute Difference value.

4 Conclusion

This study used the cross-platform LiDAR and photograph to prove the interchangeability between the data derived from different platforms in the term of performing LCC. Compared with LR-DI and LiDAR-I, Image Fusion approach improved the performance of cross-platform LCC. And all the approaches have the results of over 0.65 OA and around 0.6 Macro-F1 shown in Figure 2. To put it another way, to some content, cross-platform data can be used for inter-predicting each other. Noteworthy, digital imagery solely is not sufficient for producing TL mapping under multiple weather conditions because of the sensitive camera sensor in the different weathers. Responsibly, at that moment, LiDAR becomes a considerable reference in assisting to recognize the targets.

Reference

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- [2] PAN, S., YOSHIDA, K., KOJIMA, T., & NISHIYAMA, S. (2022). Drone-LiDAR-assisted Image Fusion methodology for Deep Learning-based Land Cover Classification. *Artificial Intelligence and Data Science*, 3(3), 15-25.